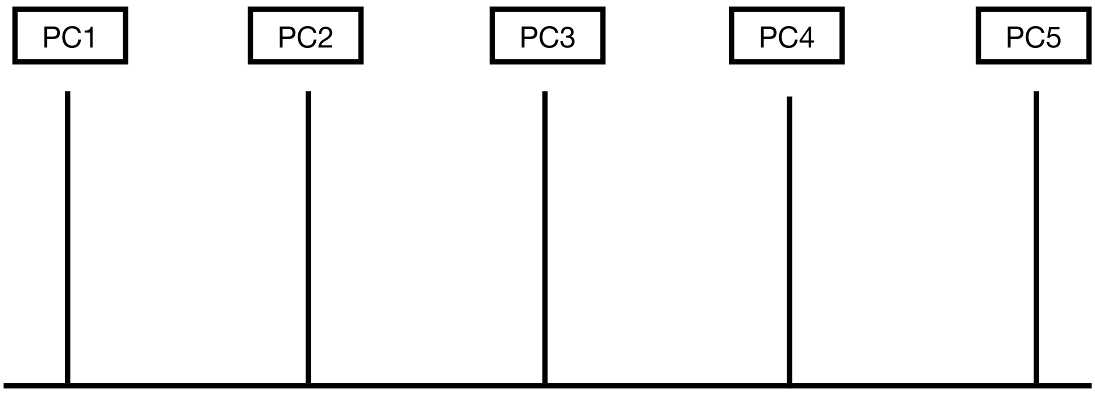
Module 1: Network + Network Fundamentals and building networks

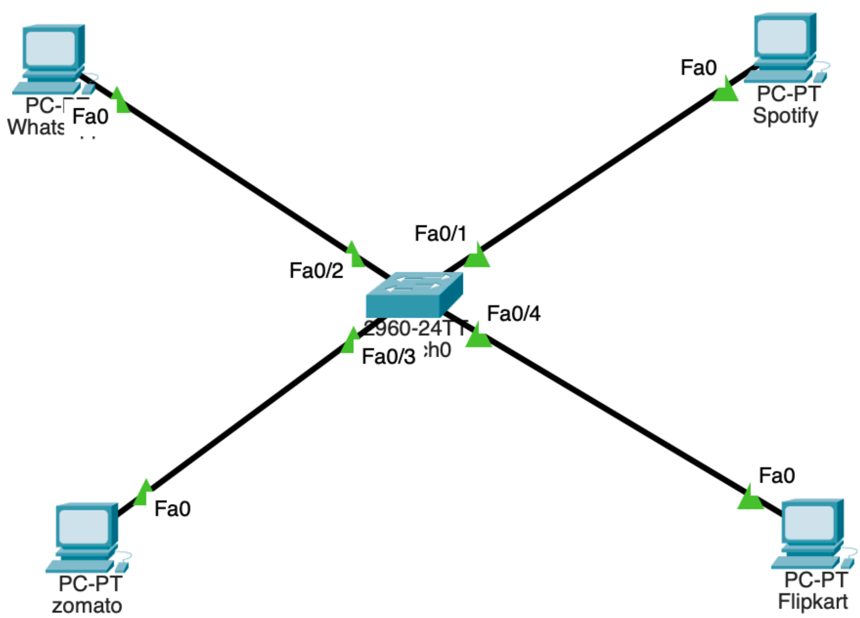
Day 2 Network Topologies

Task 1:

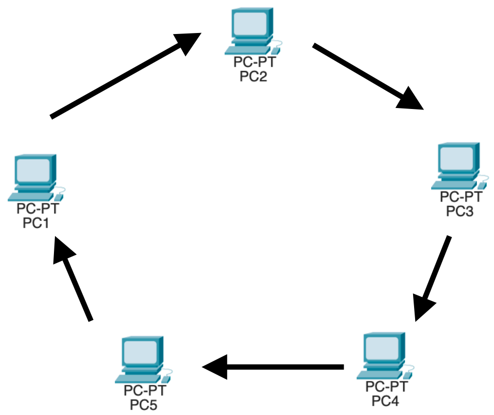


Real-world example: Bus topology was commonly used in early small office Ethernet networks using coaxial cable (10BASE2/10BASE5), where all computers tapped into one shared cable. It's rarely used today, but a modern analogy would be an old-style computer lab with a single shared cable running along the wall connecting a row of desktop PCs — cheap to set up, but if the main cable breaks anywhere, the whole network goes down.

Task 2:



Task 3:



How data travels in a ring topology: Each device is connected to exactly two neighbors, forming a closed loop, and **data travels in one direction** (as shown by the arrows) from device to device. Every device in between the sender and receiver checks the data as it passes through — if the address doesn't match, it simply forwards the data on to the next device in the ring, repeating until it reaches its actual destination.

Task 4:

Tolology	Advantage	Disadvantage
Mesh	1. Highly reliable — multiple paths mean one failed link doesn't disconnect the network 2. No single point of failure since every device connects directly to every other device	1. Very expensive and complex to wire — cabling grows rapidly as devices increase 2. Difficult to install and maintain, especially at scale

Tolology	Advantage	Disadvantage
Star	1. Easy to install and manage — just connect each device to the central switch/hub 2. If one device's cable fails, only that device is affected, not the whole network	1. If the central switch/hub fails, the entire network goes down 2. Requires more cable than some topologies since every device needs its own run back to the center